

STD 1: Statistical Analysis (Std 1)

S3 Bivariate Data Analysis (Y12)

Teacher: Denis Vlismas

Exam Equivalent Time: 73.5 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

IMPORTANT FEATURES AND TIPS FROM EXAM HISTORY

- *MS-S3 Further Statistical Analysis* has contributed 9% and 4% to the 2020 and 2019 Std1 exams respectively.
- It is a Year 12 topic that includes content found in Std2's *MS-S4 Bivariate Data Analysis*, but notably removes any analysis of *Pearson's Correlation Co-Efficient*.

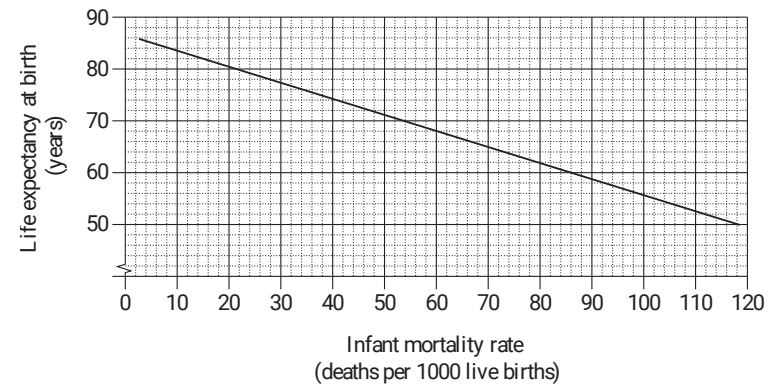
ANALYSIS - What to Expect and Common pitfalls

- *Further Statistical Analysis* investigates scatterplots, correlation, lines of best fit, gradient interpretation etc ..
- When examined, individual questions can attract large mark allocations as it did in 2020 Std1 22 which was a multi-part question worth a full 6-marks.
- Past questions have proven challenging, with every longer answer question asked in the last 10 years producing a sub-50% mean mark in at least one part.
- A content coverage note: the Standard 1 syllabus omits any explicit reference to "interpret the intercept and gradient of a [line of best fit]", which is stated in the Standard 2 syllabus. Notwithstanding this, some database questions look at this area as it represents knowledge that comes under other Standard 1 syllabus content such as: *describe the patterns, features and associations of bivariate datasets*, and *recognise the limitations of interpolation and extrapolation*.
- **Pitfalls:** marker's comments have highlighted that students have shown a lack of understanding in finding equations of best fit, interpreting gradients and identifying limitations of an equation - all areas well covered in this question database.

Questions

1. Algebra, STD2 A2 2017 HSC 3 MC

The graph shows the relationship between infant mortality rate (deaths per 1000 live births) and life expectancy at birth (in years) for different countries.

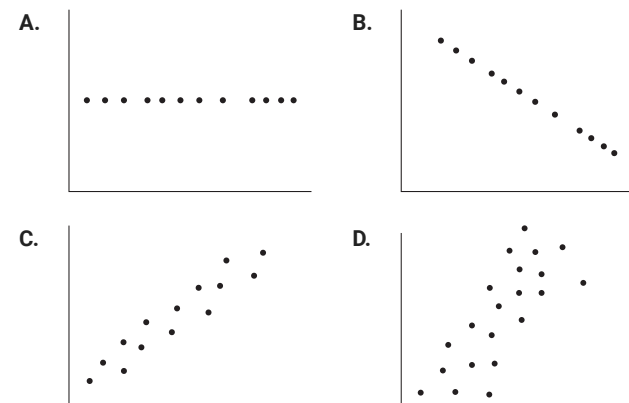


What is the life expectancy at birth in a country which has an infant mortality rate of 60?

- A. 68 years
- B. 69 years
- C. 86 years
- D. 88 years

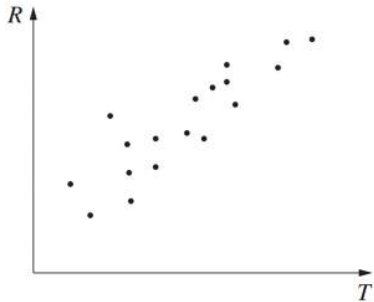
2. Statistics, STD2 S4 2017 HSC 12 MC

Which of the data sets graphed below has the largest positive correlation coefficient value?



3. Statistics, STD2 S4 2008 HSC 12 MC

A scatterplot is shown.



Which of the following best describes the correlation between *R* and *T*?

- (A) Positive
- (B) Negative
- (C) Positively skewed
- (D) Negatively skewed

4. Statistics, STD2 S4 2007 HSC 9 MC

Which of the following would be most likely to have a positive correlation?

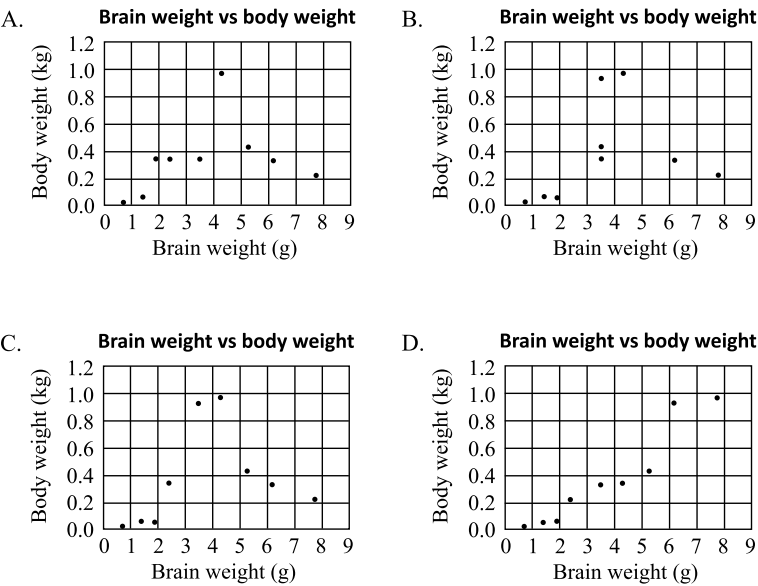
- (A) The population of a town and the number of schools in that town
- (B) The price of petrol per litre and the number of litres of petrol sold
- (C) The hours training for a marathon and the time taken to complete the marathon
- (D) The number of dogs per household and the number of televisions per household

5. Statistics, STD1 S3 2020 HSC 4 MC

The table shows the average brain weight (in grams) and average body weight (in kilograms) of nine different mammals.

<i>Brain weight (g)</i>	0.7	0.4	1.9	2.4	3.5	4.3	5.3	6.2	7.8
<i>Body weight (kg)</i>	0.02	0.06	0.05	0.04	0.93	0.97	0.43	0.33	0.22

Which of the following is the correct scatterplot for this dataset?



6. Statistics, STD2 S4 2012 HSC 11 MC

Which of the following relationships would most likely show a negative correlation?

- (A) The population of a town and the number of hospitals in that town.
- (B) The hours spent training for a race and the time taken to complete the race.
- (C) The price per litre of petrol and the number of people riding bicycles to work.
- (D) The number of pets per household and the number of computers per household.

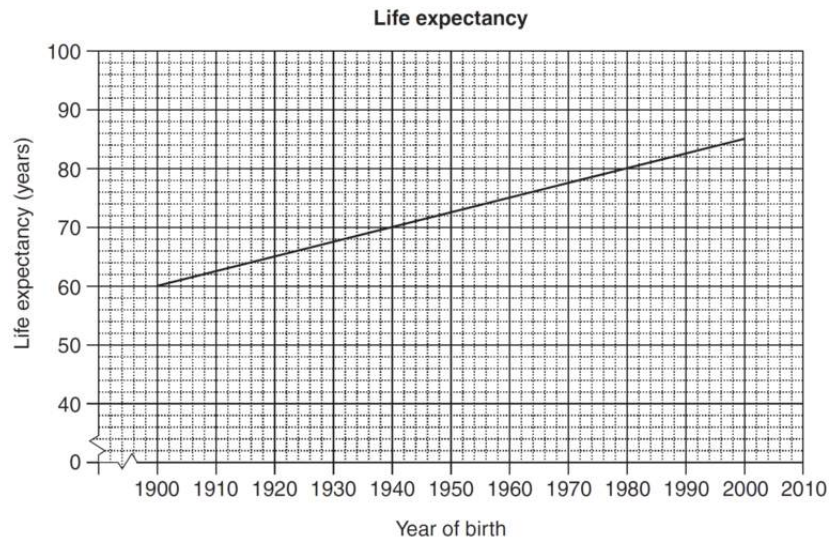
7. Statistics, STD2 S4 EQ-Bank 2

Pedro is planning a statistical investigation.

List the steps that Pedro must follow to execute the statistical investigation correctly. (2 marks)

8. Algebra, STD2 A2 2016 HSC 29e

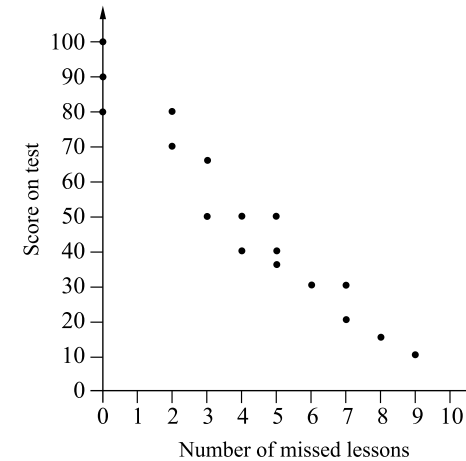
The graph shows the life expectancy of people born between 1900 and 2000.



- According to the graph, what is the life expectancy of a person born in 1932? **(1 mark)**
- With reference to the value of the gradient, explain the meaning of the gradient in this context. **(2 marks)**

9. Statistics, STD1 S3 2020 HSC 22

A group of students sat a test at the end of term. The number of lessons each student missed during the term and their score on the test are shown on the scatterplot.



- Describe the strength and direction of the linear association observed in this dataset. **(2 marks)**
- Calculate the range of the test scores for the students who missed no lessons. **(1 mark)**
- Draw a line of the best fit in the scatterplot above. **(1 mark)**
- Meg did not sit the test. She missed five lessons.
Use the line of the best fit drawn in part (c) to estimate Meg's score on this test. **(1 mark)**
- John also did not sit the test and he missed 16 lessons.
Is it appropriate to use the line of the best fit to estimate his score on the test? Briefly explain your answer. **(1 mark)**

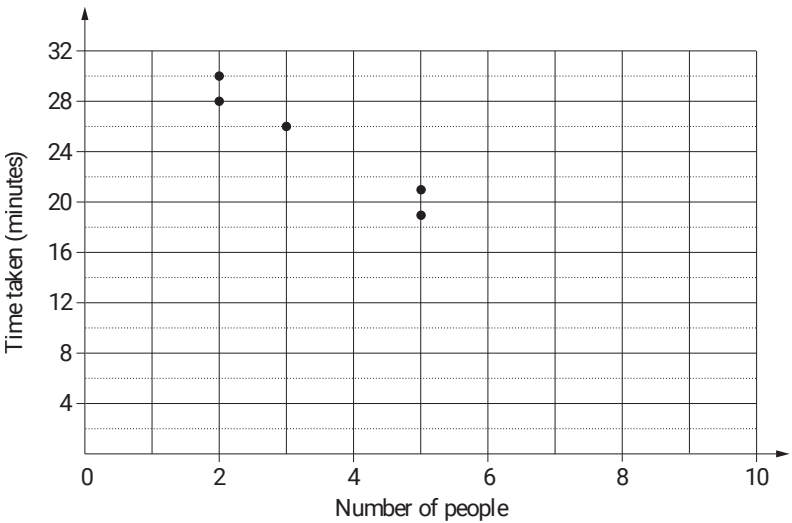
10. Statistics, STD1 S3 2021 HSC 18

People are placed into groups to complete a puzzle. There are 9 different groups.

The table shows the number of people in each group and the amount of time, in minutes, for each group to complete the puzzle.

Number of people	2	2	3	5	5	7	7	7	8
Time taken (minutes)	28	30	26	19	21	12	13	11	8

a. Complete the scatterplot by adding the last four points from the table. (2 marks)



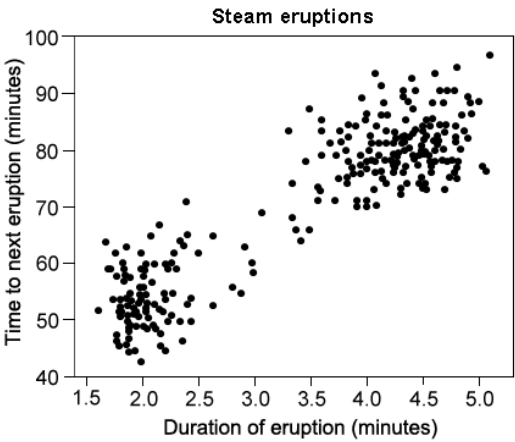
- b. Add a line of best fit by eye to the graph in part (a). (1 mark)
- c. The graph in part (a) shows the association between the time to complete the puzzle and the number of people in the group.
- Identify the form (linear or non-linear), the direction and the strength of the association. (3 marks)
- d. Calculate the mean of the time taken to complete the puzzle for the three groups of size 7 observed in the dataset. (1 mark)

11. Statistics, STD2 S3 2012 HSC 29a*

Tourists visit a park where steam erupts from a particular geyser.

The brochure for the park has a graph of the data collected for this geyser over a period of time.

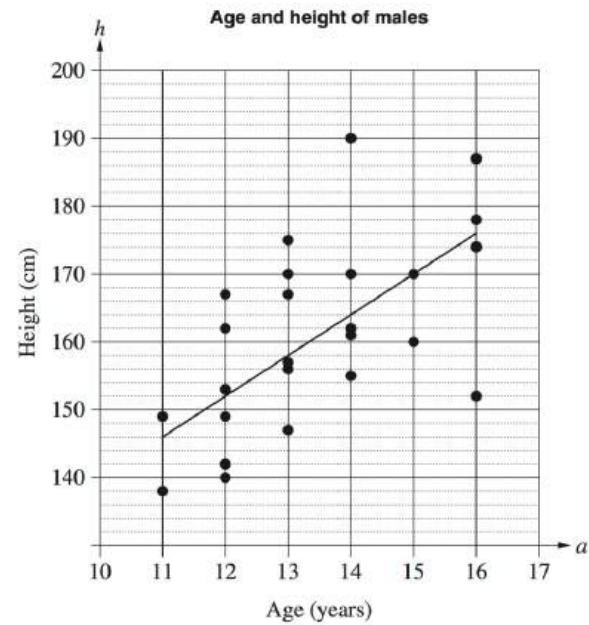
The graph shows the duration of an eruption and the time until the next eruption, timed from the end of one eruption to the beginning of the next.



- i. Tony sees an eruption that lasts 4 minutes. Based on the data in the graph, what is the minimum time that he can expect to wait for the next eruption? (1 mark)
- ii. Julia saw two consecutive eruptions, one hour apart. Based on the data in the graph, what was the longest possible duration of the first eruption that she saw? (1 mark)
- iii. What does the graph suggest about the association between the duration of an eruption and the time to the next eruption? (1 mark)

12. Statistics, STD2 S4 2013 HSC 28b

Ahmed collected data on the age (a) and height (h) of males aged 11 to 16 years. He created a scatterplot of the data and constructed a line of best fit to model the relationship between the age and height of males.



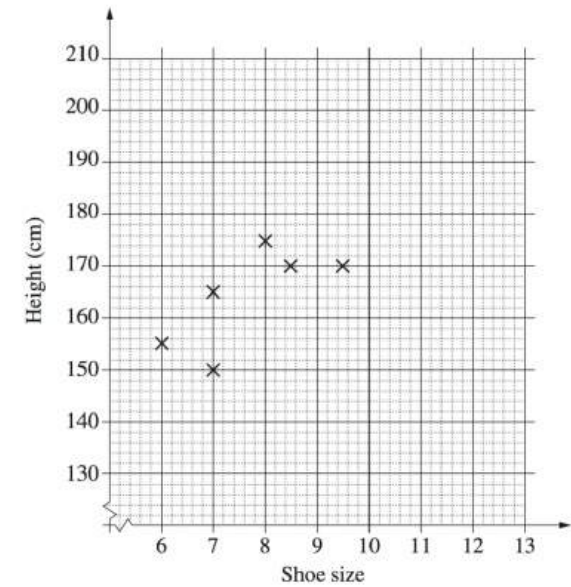
- i. Determine the gradient of the line of best fit shown on the graph. (1 mark)
- ii. Explain the meaning of the gradient in the context of the data. (1 mark)
- iii. Determine the equation of the line of best fit shown on the graph. (2 marks)
- iv. Use the line of best fit to predict the height of a typical 17-year-old male. (1 mark)
- v. Why would this model not be useful for predicting the height of a typical 45-year-old male? (1 mark)

13. Statistics, STD2 S4 2015 HSC 28e

The shoe size and height of ten students were recorded.

Shoe size	6	7	7	8	8.5	9.5	10	11	12	12
Height	155	150	165	175	170	170	190	185	200	195

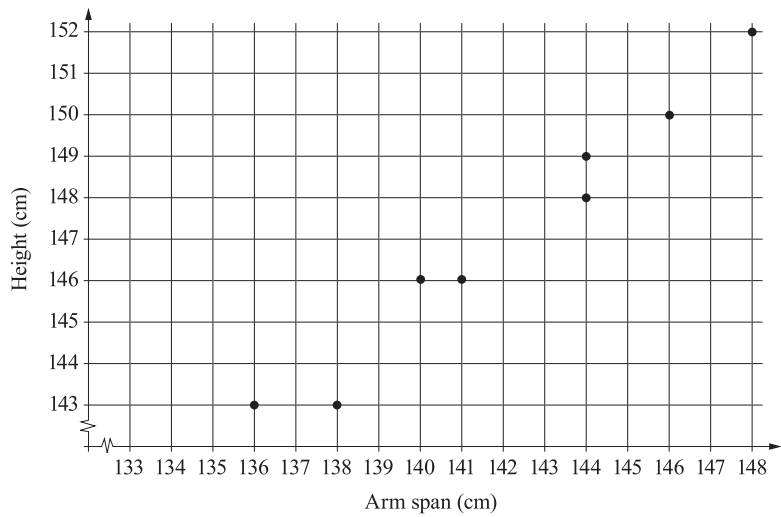
i. Complete the scatter plot AND draw a line of fit by eye. (2 marks)



- ii. Use the line of fit to estimate the height difference between a student who wears a size 7.5 shoe and one who wears a size 9 shoe. (1 mark)
- iii. A student calculated the correlation coefficient to be 1 for this set of data. Explain why this cannot be correct. (1 mark)

14. Statistics, STD1 S3 2019 HSC 27

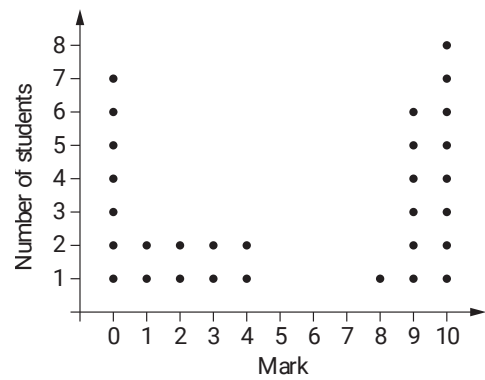
A set of bivariate data is collected by measuring the height and arm span of eight children. The graph shows a scatterplot of these measurements.



- a. On the graph, draw a line of best fit by eye. (1 mark)
- b. Robert is a child from the class who was absent when the measurements were taken. He has an arm span of 147 cm. Using your line of best fit from part (a), estimate Robert's height. (1 mark)

15. Statistics, STD2 S3 2017 HSC 29d*

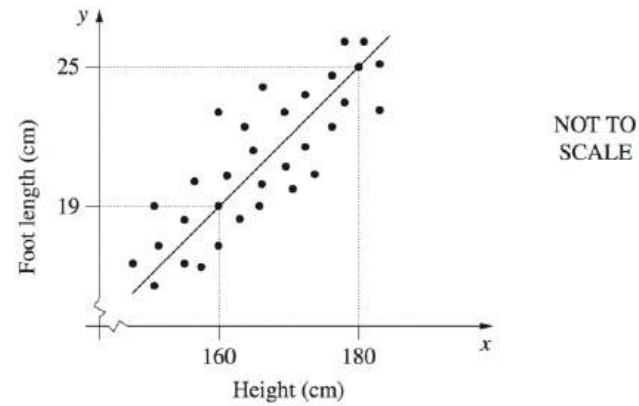
All the students in a class of 30 did a test.
The marks, out of 10, are shown in the dot plot.



- i. Find the median test mark. (1 mark)
- ii. The mean test mark is 5.4. The standard deviation of the test marks is 4.22.
Using the dot plot, calculate the percentage of the marks which lie within one standard deviation of the mean. (2 marks)

16. Statistics, STD2 S4 2006 HSC 27b

Each member of a group of males had his height and foot length measured and recorded. The results were graphed and a line of fit drawn.



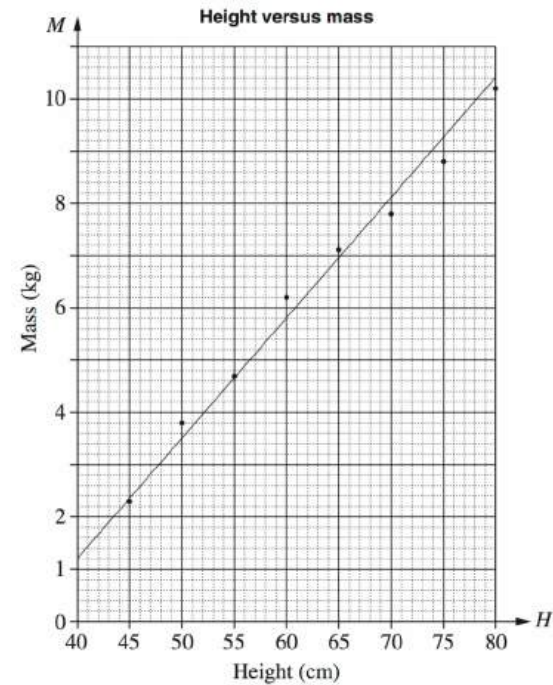
- i. Why does the value of the y -intercept have no meaning in this situation? (1 mark)
- ii. George is 10 cm taller than his brother Harry. Use the line of fit to estimate the difference in their foot lengths. (1 mark)
- iii. Sam calculated a correlation coefficient of -1.2 for the data. Give TWO reasons why Sam must be incorrect. (2 marks)

17. Statistics, STD2 S4 2009 HSC 28b

The height and mass of a child are measured and recorded over its first two years.

Height (cm), H	45	50	55	60	65	70	75	80
Mass (kg), M	2.3	3.8	4.7	6.2	7.1	7.8	8.8	10.2

This information is displayed in a scatter graph.

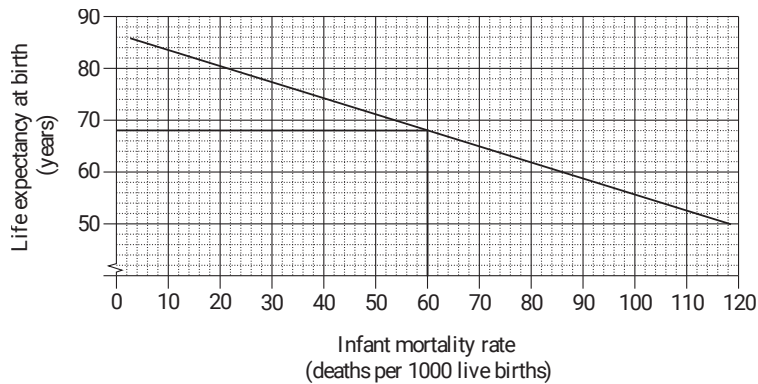


- i. Describe the correlation between the height and mass of this child, as shown in the graph. (1 mark)
- ii. A line of best fit has been drawn on the graph.
Find the equation of this line. (2 marks)

Worked Solutions

1. Algebra, STD2 A2 2017 HSC 3 MC

When infant mortality rate is 60, life expectancy at birth is 68 years (see below).



⇒ A

2. Statistics, STD2 S4 2017 HSC 12 MC

Largest positive correlation occurs when both variables move in tandem. The tighter the linear relationship, the higher the correlation.

⇒ C

(Note that B is negatively correlated)

3. Statistics, STD2 S4 2008 HSC 12 MC

Correlation is positive.

NB. The skew does not refer to correlation.

⇒ A

4. Statistics, STD2 S4 2007 HSC 9 MC

Positive correlation means that as one variable increases, the other tends to increase also.

⇒ A

5. Statistics, STD1 S3 2020 HSC 4 MC

Consider data point (1.9, 0.05)

→ Eliminate A (too high)

→ Eliminate D (should be below 2nd data point)

Consider data point (2.4, 0.34)

→ Eliminate B (not on graph)

⇒ C

6. Statistics, STD2 S4 2012 HSC 11 MC

Increased hours training should reduce the time to complete a race.

♦ Mean mark 43%

⇒ B

7. Statistics, STD2 S4 EQ-Bank 2

Steps are:

- Identify a problem and pose a statistical question
- Collect or obtain data
- Represent and analyse the data collected or obtained
- Communicate and interpret findings.

8. Algebra, STD2 A2 2016 HSC 29e

i. 68 years

ii. Using (1900,60), (1980,80)

$$\begin{aligned}\text{Gradient} &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{80 - 60}{1980 - 1900} \\ &= 0.25\end{aligned}$$

♦♦ Mean mark part (ii) 33%

∴ After 1900, life expectancy increases by 0.25 years for each year later that someone is born.

9. Statistics, STD1 S3 2020 HSC 22

a. **Strength : strong**

Direction : negative

◆ Mean mark part (a) 45%.

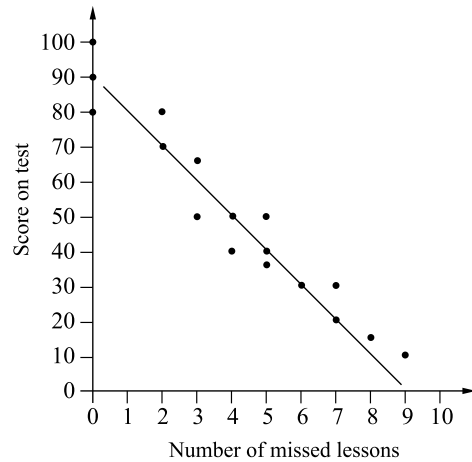
b. **Range = high – low**

$$= 100 - 80$$

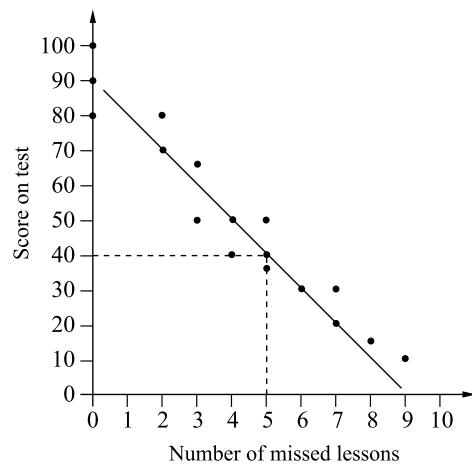
$$= 20$$

◆◆ Mean mark part (b) 31%.

c.



d.



∴ **Meg's estimated score = 40**

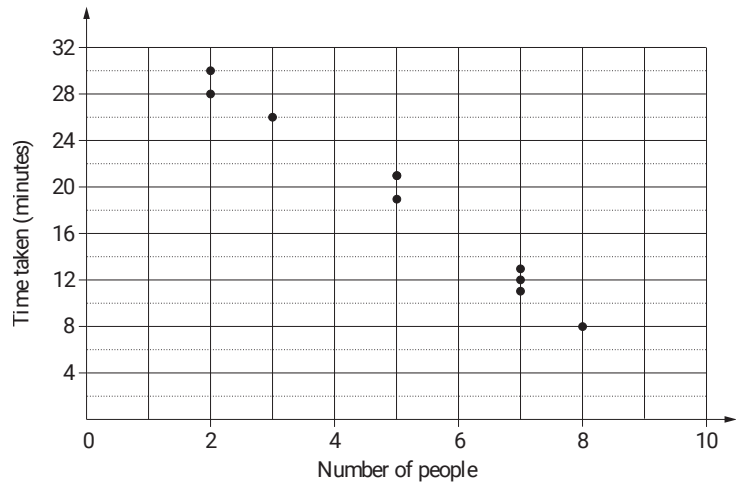
e. **John's missed days are too extreme and the LOBF is not appropriate.**

◆ Mean mark part (e) 38%.

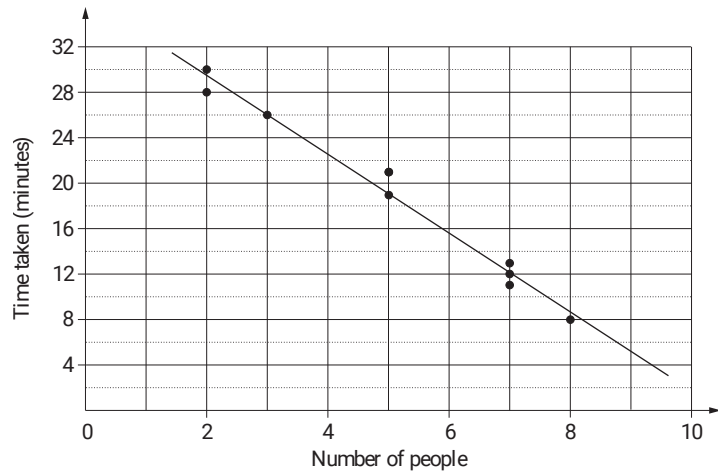
The model would estimate a negative score for John which is impossible.

10. Statistics, STD1 S3 2021 HSC 18

a.



b.



c. The association is linear, negative and strong.

♦ Mean mark part (c) 50%.

d. Mean time (7 people) = $\frac{12 + 13 + 11}{3}$
= 12 minutes

11. Statistics, STD2 S3 2012 HSC 29a*

i. 70 minutes (refer to graph)

ii. 3 minutes

Locate 60 minutes on y -axis and look for longest duration (i.e. the largest x -axis value) for that given y -value.

♦ Mean mark 47%

iii. It suggests that the longer an eruption lasts for, the longer you will wait until the next one, or there is a positive correlation between the length of an eruption and the time until the next one.

12. Statistics, STD2 S4 2013 HSC 28b

$$\begin{aligned} \text{i. Gradient} &= \frac{\text{rise}}{\text{run}} \\ &= \frac{176 - 146}{16 - 11} \\ &= \frac{30}{5} \\ &= 6 \end{aligned}$$

ii. Males should grow 6cm per year
between the ages 11-16.

$$\begin{aligned} \text{iii. Gradient} &= 6 \\ \text{Passes through } (11, 146) \\ y - y_1 &= m(x - x_1) \\ h - 146 &= 6(a - 11) \\ &= 6a - 66 \\ \therefore h &= 6a + 80 \end{aligned}$$

$$\begin{aligned} \text{iv. If } a &= 17 \\ h &= (6 \times 17) + 80 \\ &= 182 \end{aligned}$$

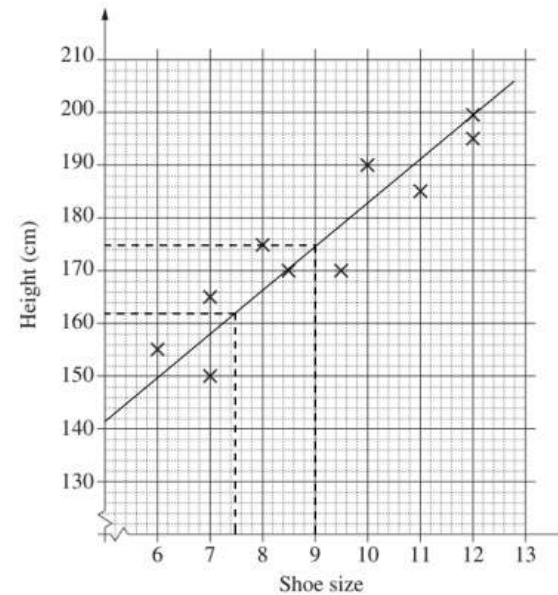
$\therefore$ A typical 17 year old is expected to be 182cm

v. People slow and eventually stop growing
after they become adults.

♦♦ Mean marks of 38%, 26% and 25% respectively for parts (i)-(iii).
MARKER'S COMMENT: Interpreting gradients has been consistently examined in recent history and almost always poorly answered.

13. Statistics, STD2 S4 2015 HSC 28e

i.



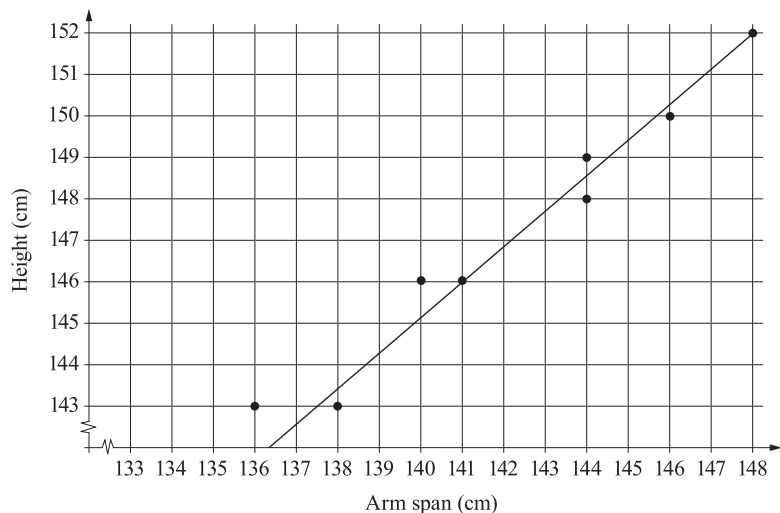
ii. Shoe size $7\frac{1}{2}$ gives a height estimate
of 162 cm (see graph)
Shoe size 9 gives a height estimate
of 175 cm (see graph)
 $\therefore$ Height difference = $175 - 162$
 $= 13$ cm

iii. A correlation co-efficient of 1 would mean
that all data points occur on the line of best
fit which clearly isn't the case.

♦ Mean mark 39%.

14. Statistics, STD1 S3 2019 HSC 27

a.



b. Robert's height ≈ 151.1 cm

♦ Mean mark part (a) 38%.

(Answers can vary slightly depending on line of best fit drawn).

15. Statistics, STD2 S3 2017 HSC 29d*

i. Median = $\frac{15\text{th} + 16\text{th score}}{2}$

$$= \frac{4 + 8}{2}$$

$$= 6$$

♦ Mean mark 50%.

ii. Lower limit = $5.4 - 4.22 = 1.18$
 Upper limit = $5.4 + 4.22 = 9.62$

♦♦ Mean mark 34%.

$\therefore$ Percentage in between

$$= \frac{13}{30} \times 100$$

$$= 43.33\ldots$$

$$= 43\% \text{ (nearest \%)}$$

16. Statistics, STD2 S4 2006 HSC 27b

- The y-intercept occurs when $x = 0$. It has no meaning to have a height of 0 cm.
- A 20 cm height difference results in a foot length difference of 6 cm.

 $\therefore$ A 10 cm height difference means George should have a 3 cm longer foot.
- A correlation co-efficient must be between -1 and 1 .
 Foot length is positively correlated to a person's height and therefore isn't a negative value.

17. Statistics, STD2 S4 2009 HSC 28b

- The correlation between height and mass is positive and strong.

♦ Mean mark 48%.

- Using $P_1(40, 1.2)$ and $P_2(80, 10.4)$

$$\text{Gradient} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{10.4 - 1.2}{80 - 40}$$

$$= \frac{9.2}{40}$$

$$= 0.23$$

Line passes through $P_1(40, 1.2)$

Using $y - y_1 = m(x - x_1)$

$$y - 1.2 = 0.23(x - 40)$$

$$y - 1.2 = 0.23x - 9.2$$

$$y = 0.23x - 8$$

$\therefore$ Equation of the line is $M = 0.23H - 8$

♦♦♦ Mean mark 18%.
MARKER'S COMMENT: Many students had difficulty due to the fact the horizontal axis started at $H = 40\text{cm}$ and not the origin.